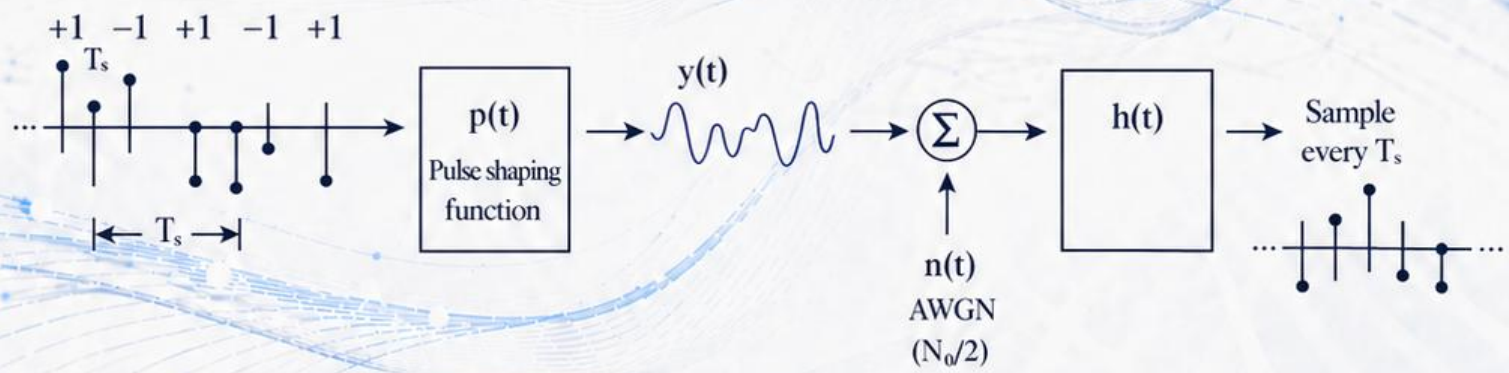
Cairo University
Faculty of Engineering

Introduction to Communications

PROJECT # 3



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INTRODUCTION TO COMMUNICATIONS PROJECT #3

1. Introduction and Project Overview

This project evaluates fundamental digital communication components using MATLAB. The analysis covers optimal receiver design (Matched Filters and Correlators), Bit Error Rate (BER) performance in AWGN channels, and the mitigation of Inter-Symbol Interference (ISI) using Square-Root Raised Cosine (SRRC) filtering.

2. Theoretical Background

2.1 PAM Signaling and Pulse Shaping

In binary PAM, bits 0,1 are mapped to bipolar symbols a_k ∈ {+1, -1} ,The transmitted signal y[n] is the convolution of an upsampled impulse train with a pulse shape p[n]. This project uses a normalized pulse p[n] = [5, 4, 3, 2, 1] / sqrt{55}, ensuring unit energy (E_b = 1).

2.2 Optimal Receiver Design

- **Matched Filter (MF):** Maximizes Signal-to-Noise Ratio (SNR) at the sampling instant. The impulse response is the time-reversed pulse: h[n] = p[N-1-n].
- **Correlator:** An equivalent implementation that multiplies the received signal by p[n] and integrates over Ts.
- **Theoretical BER:** For binary antipodal signaling in AWGN

2.3 ISI and Nyquist Criterion

ISI occurs when pulse energy spills into adjacent symbol periods. To eliminate ISI, the overall system response $h_{overall}(t)$ must satisfy the Nyquist criterion: $h_{overall}(kTs) = 0$ for k != 0. This is achieved using **Raised Cosine (RC)** filters, often split into two **Square-Root Raised Cosine (SRRC)** filters at the Tx and Rx

3. Task 1 — Matched Filters and Correlators

3.1 Signal Generation

This section walks through each step of the signal generation chain, as described in the project specification and implemented in the MATLAB code.

Step 1 — Generate 10 Random Binary Bits

The randi function generates 10 random bits uniformly drawn from {0, 1}. This models the source of digital information to be transmitted.

```
N = 10;

bits = randi([0 1], 1,N);
```

Step 2 — Convert to Bipolar Symbols (+1 / -1)

Binary bits are mapped to bipolar symbols using the linear transformation a = 2·b - 1. This maps bit '1' to symbol +1 and bit '0' to symbol -1, which is the standard representation for binary polar (antipodal) signaling.

```
symbols = 2*bits - 1;
```

Step 3 — Upsampling (Generating Impulses Every Ts)

The MATLAB upsample(symbols, 5) command inserts 4 zeros between each symbol value. This models the impulse train representation: an impulse of magnitude +1 or -1 occurs every Ts = 1 second, with 5 samples per symbol period (sampling interval = 200 ms).

```
impulses = upsample(symbols, samples_per_sym);
```

The result is a sparse array of length 10 × 5 = 50 samples, where only every 5th sample is non-zero (equal to the symbol value), and all others are zero.

Step 4 — Pulse Shaping via Convolution

Convolving the upsampled symbol sequence with the pulse p[n] effectively places a scaled copy of p at each non-zero impulse location. The result y[n] is the discrete-time transmitted PAM signal.

```
p = [5 4 3 2 1] / sqrt(55);

y = conv(upsampled_symbols, p);
```

After convolution, y has length = length(upsampled_symbols) + length(p) - 1 = 50 + 5 - 1 = 54 samples. Each symbol contributes a pulse of shape p scaled by +1 or -1.

Step 5 — Applying the Filters

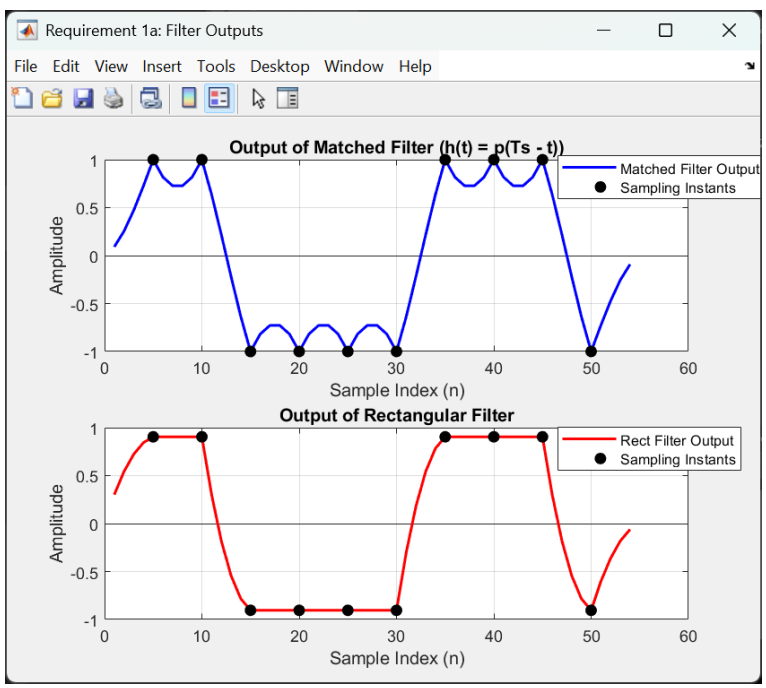
Two receiving filters are applied to y[n]:

- Matched filter: h_matched = fliplr(p) = [1 2 3 4 5]/sqrt(55). This is the time-reverse of p.

- Rectangular filter: h_rect = [1 1 1 1 1]/sqrt(5). A uniform average over one symbol period, also energy-normalized.

```
h_mf = fliplr(p);

h_rect = ones(1, samples_per_sym) / sqrt(samples_per_sym);


out_mf = conv(y, h_mf);

out_rect = conv(y, h_rect);
```

3.2 Requirement 1a: Filter Output Comparison



Results:

The two filters — matched and rectangular — are plotted on two subplots of the same figure. At the optimal sampling instants (every 5th sample after filter settling), the matched filter output has a larger magnitude than the rectangular filter. This is because the matched filter is designed to maximize SNR, which in a noise-free system corresponds to maximizing the signal energy at the sampling point relative to the input.

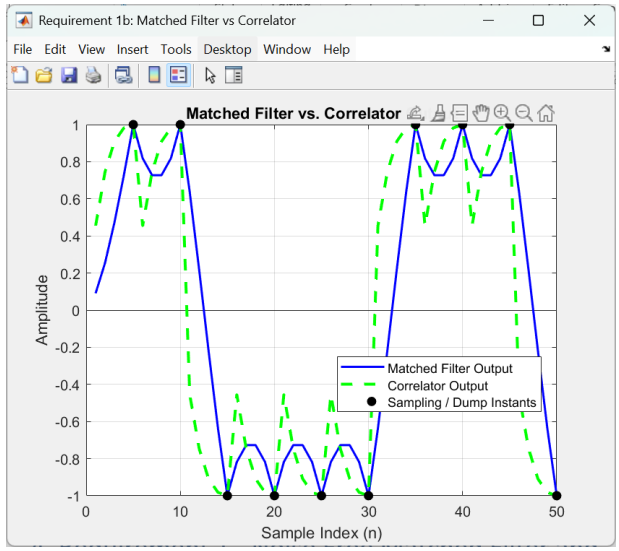
The matched filter produces peaks that are sharper and better separated at the sampling instants, making symbol detection more reliable. The rectangular filter, while simpler, does not align its frequency response with the transmitted pulse spectrum, resulting in a suboptimal (lower) peak at the sampling instant.

3.2 Requirement 1b: Matched Filter vs. Correlator

The correlator is implemented using a for-loop that processes the signal in windows of 5 samples (one symbol period). For each window, it multiplies by p[n] and computes the cumulative sum (integration). The final value in each window is the 'dump' output — identical to the matched filter sample at the optimal instant.

```
out_corr = zeros(1, length(y));

for k = 1:N

  idx = (k-1)*samples_per_sym + 1 : k*samples_per_sym;

  out_corr(idx) = cumsum(y(idx) .* p);

end
```

The loop index runs from 1 to the last valid start of a 5-sample window. The cumsum function implements the discrete-time equivalent of integration: it returns a running sum that accumulates from the start of the window. At the 5th sample of each window, the cumsum output equals the full inner product of the received signal segment with p[n], which is the correlator decision variable.



Result:

The matched filter output and the correlator output are plotted on the same axes. The correlator is shown as a dashed green line (cumulative sum within each symbol period), while the matched filter is shown as a solid blue line. Black dots mark the sampling/dump instants.

At every sampling instant (every 5th sample), the correlator and matched filter outputs are identical. This confirms the theoretical equivalence: the matched filter and correlator produce the same decision statistic at the sampling instant, even though their intermediate (inter-sample) values differ. This is the fundamental reason why correlators and matched filters are used interchangeably in practice.

4. Task 2 — Noise Analysis and BER

4.1 Simulation Setup

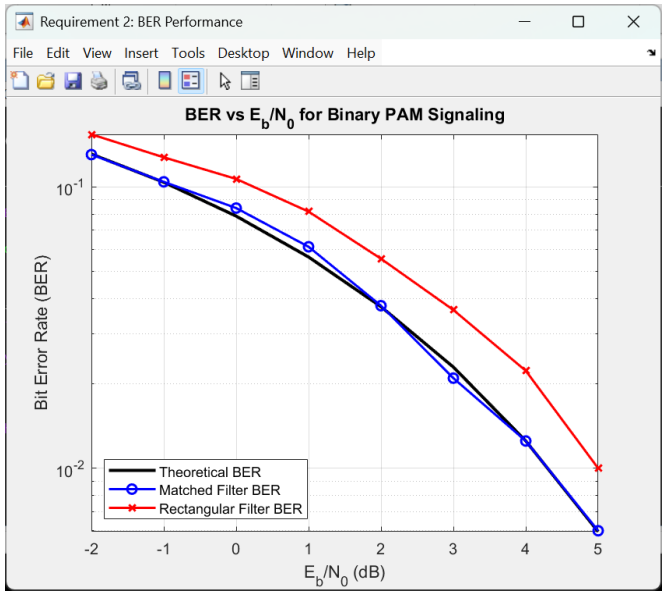
To ensure statistical accuracy, 10,000 bits are simulated across an Eb/N0 range of -2 dB to +5 dB.

Parameter	Value	Notes
Number of bits	10,000	Ensures statistically reliable BER
Upsampling factor	5	5 samples per symbol
Eb/N0 range	-2 dB to +5 dB	8 points, 1 dB step
Pulse energy (Eb)	1	Normalized pulse
Noise mean	0	Zero-mean AWGN
Noise variance	N0/2	Per AWGN channel model

4.2 Requirement 2: BER Curves

The simulation compares the Measured MF BER, Measured Rectangular BER, and the Theoretical Curve.

- **Matched Filter:** The curve closely tracks the theoretical 0.5erfc(sqrt{Eb/N0}).
- **Rectangular Filter:** Consistently exhibits a higher BER. The distance between this curve and the MF curve represents the **matching loss** due to the suboptimal filter shape.



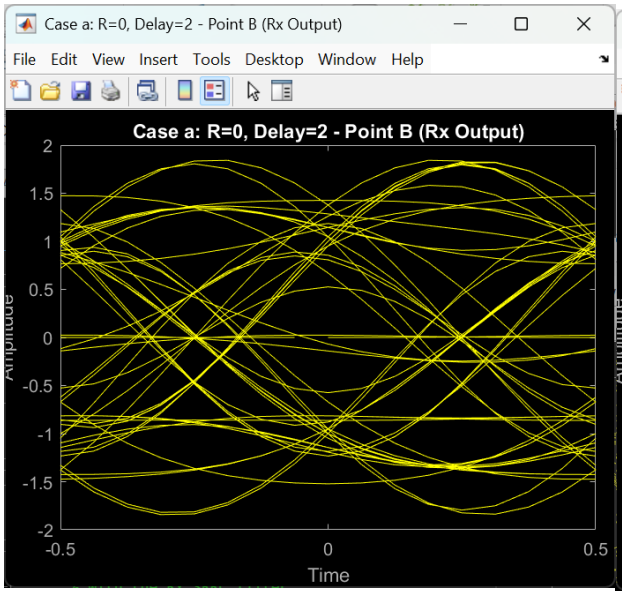
Task 3 — ISI and Raised Cosine Filters

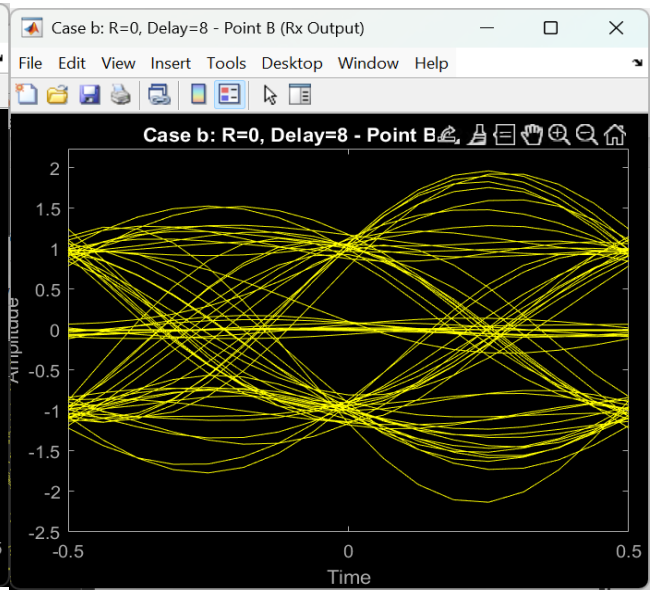
5.1 System Configuration

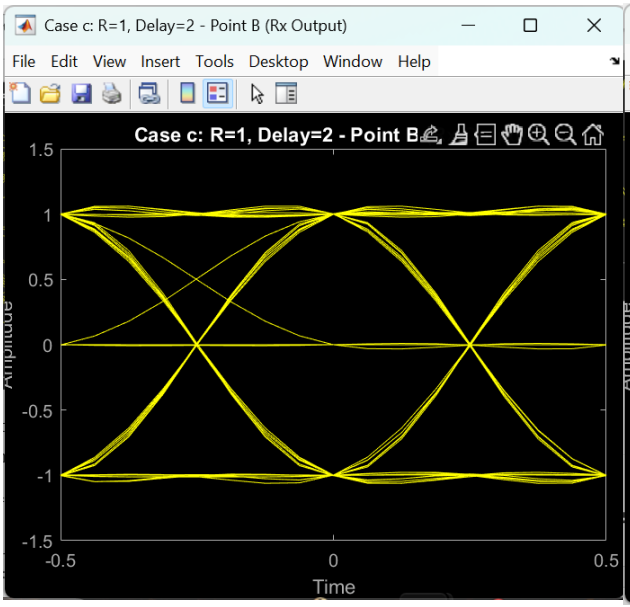
The system utilizes SRRC filters at both Tx (Point A) and Rx (Point B). The combination forms a full Raised Cosine response. Four cases are analyzed based on Roll-off (R) and Delay.

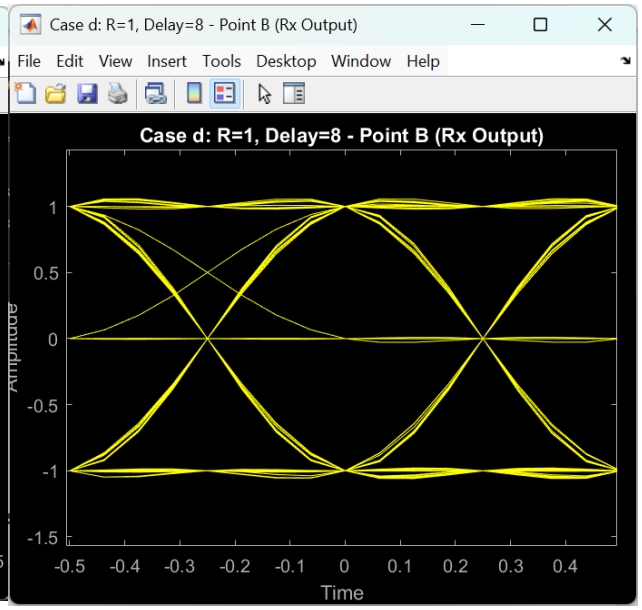
5.2 Requirement 3: Eye Diagram Analysis

Case	Roll-off R	Delay	Bandwidth	Time Truncation
a	0 (Nyquist)	2	Minimum (1/2Ts)	Short (severe)
b	0 (Nyquist)	8	Minimum (1/2Ts)	Longer (moderate)
c	1 (Maximum)	2	Double (1/Ts)	Short (mild)
d	1 (Maximum)	8	Double (1/Ts)	Longer (negligible)









The eye diagrams generated for the four test cases illustrate the critical relationship between the roll-off factor (R) and filter length (delay) in managing ISI. Below is the comparative analysis of the results observed at Point B (Receiver Output):

- **Case (a) [R=0, Delay=2]: Nearly Closed Eye**. The eye is significantly closed due to severe Inter-Symbol Interference (ISI). This occurs because the ideal R=0 pulse is a sinc function with slow decay; truncating it to only +-2 symbol periods results in a very poor approximation.

- **Case (b) [R=0, Delay=8]: Moderately Open Eye**. Increasing the filter delay to 8 captures more of the slowly decaying sinc tails, which reduces truncation error. Consequently, the eye opens considerably compared to Case (a), proving that longer filter lengths are vital for narrow-band (R=0) configurations.
- **Case (c) [R=1, Delay=2]: Open Eye**. Despite the short delay, the eye is substantially open. The fast time-domain decay associated with maximum roll-off (R=1) makes the pulse shape inherently robust to truncation.
- **Case (d) [R=1, Delay=8]: Maximum Eye Opening**. This best-case configuration demonstrates near-ideal ISI-free transmission. The eye is wide open with minimal distortion, and the optimal sampling instant at the center is clearly defined

5. Summary and Conclusions

The simulation confirms that the **Matched Filter** is essential for minimizing BER in AWGN by maximizing sampling-instant SNR. Furthermore, **Raised Cosine** signaling effectively eliminates ISI, provided the roll-off factor and filter delay are sufficient to prevent truncation errors. For an electronics engineer, balancing R (bandwidth) and Delay (complexity/latency) is critical for practical system design.

6. MATLAB Codes

❖ GITHUB REPO LINK: MATLAB SOURCE CODES